Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

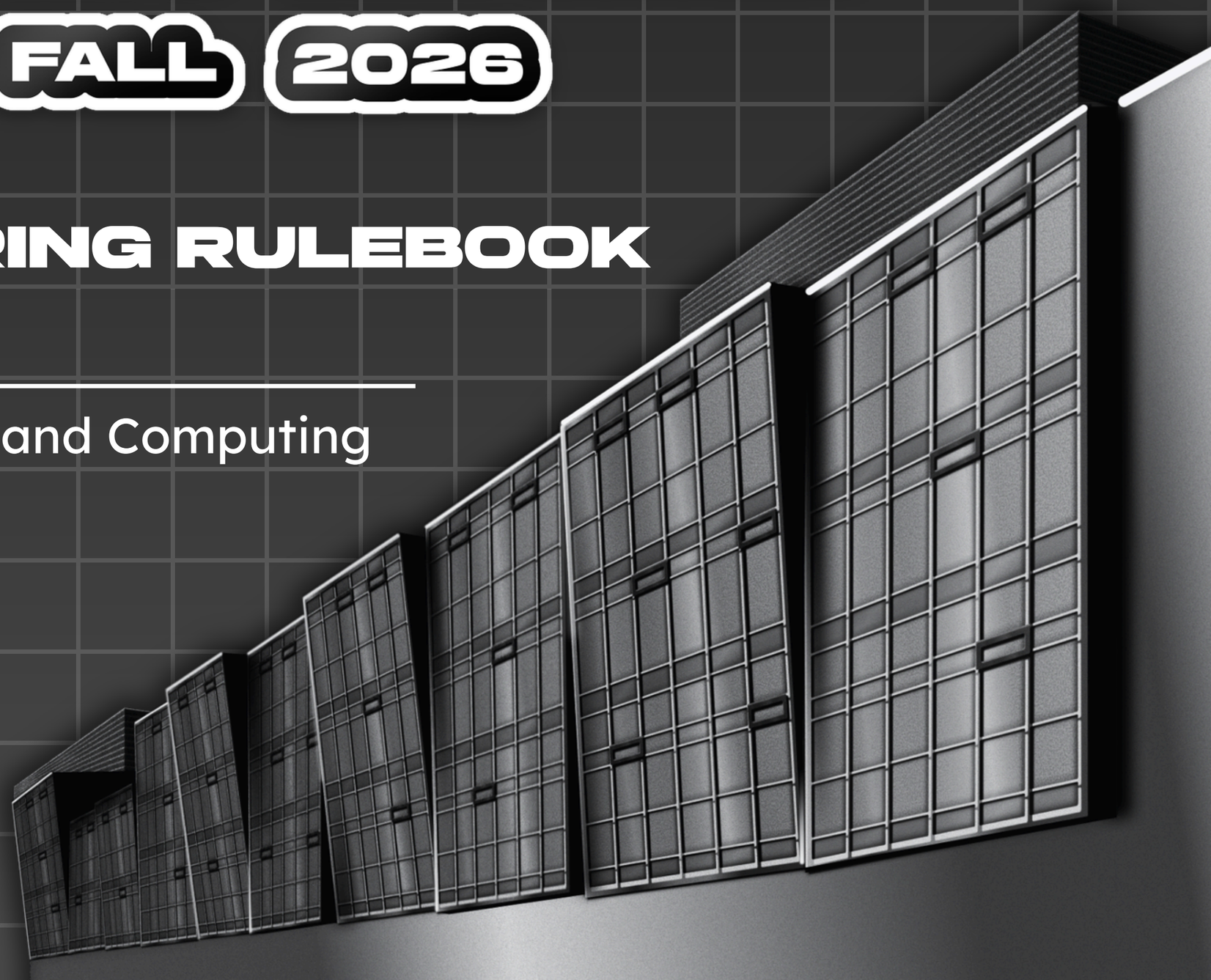
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026

BIO-ENGINEERING RULEBOOK

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



1.1. GOAL

In this competition, teams comprehensively solve a bio-engineering problem. Teams are required to identify the issues outlined in the prompt and apply a bio-engineering design process to develop a practical and feasible solution. Competitors must demonstrate an understanding of the relevant biological principles, select and justify an appropriate biological approach, and develop an engineering system that supports their proposed solution. Teams must demonstrate their solution through a non-physical prototype and technical report.

1.2. TEAM COMPOSITION

The maximum number of individuals for a Bio-Engineering team is two (2).

1.3. LANGUAGE

The official language of the MEC is English. French is recognized as a secondary language that will be accommodated upon request. At the time of registration, competitors must notify the OC that they choose to compete in French. The OC will provide all necessary documentation in French and provide live translation to make the competition fair and feasible for the competitor. All deliverables must be consistent in language. All members of a team must compete in the same language.

1.4. TOPIC

It is recommended that the topic incorporate more than one engineering discipline. However, while the topic should challenge competitors' technical knowledge and skills, it should also require competitors to evaluate the economic, environmental, political and social implications of their proposed solutions and address the requirements of the customer. Competition topics will be ones that could exist in the real world. Topics drawn from reality must be fully documented. All necessary documentation must be provided to the competitors when the problem is presented.

1.5. RESOURCES

1.5.1. Facilities Required

- One (1) amphitheater (presentation of problem and team presentations)
- One (1) workroom per team (design of solution)

1.5.2. Personnel Required

1.5.2.1. Judges

A minimum of three (3) judges (and in any excess, an odd number of judges) are required to assess the problem-solving abilities, proposed solution, inclusion of a non-physical prototype, communication skills and team dynamics of the competitors. Judges in this category should come from a variety of backgrounds including communications, sales and technical or consulting engineering experience related to the topic.

1.5.2.2. Competition Leads

The Competition Lead(s) are responsible for the design and implementation of their respective competition. The Competition Lead(s) must be present at all of the presentations. The Competition Lead(s) must present the design problem at the beginning of the competition and answer any questions raised by Competitors. The Competition Lead(s) will be available to competitors and judges during competition hours for questions and requests.

1.5.2.3. Official Timekeeper

The Official Timekeeper must be responsible for enforcing time limits during the question period, solution development, and presentations.

1.5.3. Equipment Provided by MEC

The following resources are required to be provided for Bio-Engineering teams during the competition phase:

1.5.3.1. During the Design Phase

- One (1) large workroom, subdivided with partitions for each team or one (1) separate workroom per team
- One (1) table per team
- One (1) chair per Competitor
- One (1) of a whiteboard, chart paper or blackboard
- Paper and pencils/pens for writing
- Internet connectivity
- Power source for computers

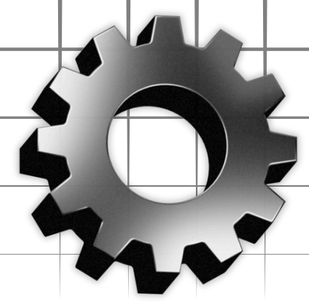
1.5.3.2. During the Presentation

- One (1) room
- One (1) computer containing the team's presentation file, report, and non-physical prototype
- One (1) digital projector; and
- Whiteboard(s) and/or blackboard(s) and associated writing implements

1.5.4. External Resources Allowed

The competitors can bring the following equipment with them:

- Any background research conducted by team members prior to the competition, including their submitted pre-competition literature review
- Any electronic material stored on CDs, USB keys, disks or other storage devices
- Please check with the organizers to make sure the format of your electronic information will be accessible using the computers provided by MEC
- Any textbooks, course notes or other reference material
- Each team member is allowed one computer



Competitors are not permitted to bring prebuilt physical prototypes or previously developed competition-specific engineering designs. All proposed engineering solutions and non-physical prototypes must be developed during the competition design period. Acceptable non-physical engineering prototypes or design representations may include CAD models, engineering drawings or sketches, circuit schematics, PCB designs, system or block diagrams, simulations, process diagrams, system, software applications, websites, user-interface prototypes, digital models, or other appropriate non-physical engineering representations.

Note: Since the use of the Internet and other external resources is permitted in this competition, all information used by competitors must be referenced carefully. Competitors are not permitted to submit work completed by anyone other than the members of their team. If there is any evidence that competitors are submitting plagiarized work, the entire team will be eliminated from the competition and their home schools will be notified. Volunteers will monitor each team during the design process to deter teams from cheating, however, competitors are expected to act in good faith with the spirit of the competition.

1.6. PROCEDURE / TIMELINE

1.6.1. Case Structure

The competition will consist of one (1) written case that will prompt the competitors to apply the bio-engineering process and propose a solution. Competitors will receive a broad research topic six (6) days prior to the competition to prepare a literature review. The complete competition case, including the specific problem, requirements, constraints and deliverables, will only be distributed on Day 1 of MEC at the beginning of the competition design period. Its solution will be required by the end of the competition design period.

1.6.2 Pre-Competition

Prior to the competition, information about the rules of the competition will be available on the competition website, metengcomp.com.

Six (6) days before the competition, registered teams will receive a broad biological research topic related to the competition. Teams will be required to conduct research and submit a pre-competition literature review based on the provided topic.

The literature review must meet the following requirements:

- Maximum length of three (3) pages, excluding references
- Minimum of three (3) peer-reviewed academic articles
- All external sources must be cited using APA format
- One (1) submission per two (2) person team
- Submission in PDF format through the designated submission form
- Submission before November 6th, 2026, at 11:59PM (Eastern time)

Competitors may only collaborate with their registered two (2) person team during the preparation period.

1.6.3 Competition

1.6.3.1. Presentation of Problem

The problem must be presented to all competitors at the beginning of the competition. The Competition Lead(s) must provide detailed explanations of what is expected from the competitors, both orally and in writing. The presentation of the problem will take no longer than fifteen (15) minutes.

1.6.3.2. Question Period

Competitors have fifteen (15) minutes to ask the Competition Leads any questions. Only the time used to ask the questions (not the answers) should be counted in the fifteen (15) minutes. Time used by judges to pose questions will not be counted.

1.6.3.3. Solution Development

Teams will be given eight (8) hours to develop their solutions to the case, produce all required deliverables, and prepare their presentations. All deliverables shall be submitted to the Competition Lead before the end of the allotted time. Competitors may finish before the end of the allotted time. All competition-specific engineering design work must be completed during the allotted design period.

Teams must also develop a non-physical engineering prototype or design representation that demonstrates how their proposed system functions, how its components interact, and how the solution could realistically be implemented. Physical Prototypes are not permitted.

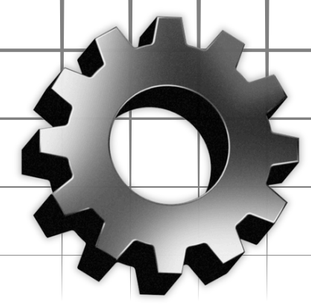
Teams are required to prepare a written technical report, with a maximum length of six (6) pages, documenting the proposed solution, relevant biological research, engineering design, technical analysis, and justification of design decisions. The report must also address problem analysis, the selected biological approach, alternative solutions and trade-offs, prototype, proposed testing and validation methods, implementation, costs, safety, environmental and social considerations, limitations, and scalability. Relevant figures and properly cited references must be included.

1.6.3.4. Rest Period

Competitors must be allowed a minimum of one (1) hour to rest before the presentation phase starts.

1.6.3.5. Solution Presentations

The presentation should cover the completed case and should present the material included in the report and any additional information and explanations required to meet the evaluation criteria below. Competitors will have a maximum of ten (10) minutes topresent their solutions. All team members must be present and participate in the presentationor be penalized by the judges. Judges then have a maximum of five (5) minutes to askquestions. Judges can ask a question at any time during the presentation. The clock must bestopped during these interruptions.



Within that period. teams must demonstrate their proposed engineering solution through an appropriate non-physical prototype or design representation during the presentation. The demonstration should clearly communicate the proposed system's functionality and the integration of biological and engineering components.

In order to ensure that all competitors cease to work on the case solutions once the design time has ended, the competitors cannot include any material in their oral presentation which is not included within their submitted written reports or presentation materials.

1.6.4. Timekeeping

The following rules will be adhered to with respect to timekeeping.

1.6.4.1. During the Design Phase

- The design period will begin immediately upon the conclusion of the question period.
- The remaining time must be announced three (3) hours, one (1) hour, thirty (30) minutes, and ten (10) minutes before the end of the allotted eight (8) hours design time.

1.6.4.2. During the Presentation

- Time is halted when a judge asks a question and when a team member answers the question during the presentation.
- The remaining time must be indicated to the competitors ten (10) minutes, five (5) minutes and one (1) minute before the end of the allotted presentation time.
- A visual countdown must be given to the presenters during the last thirty (30) seconds of the allotted presentation time.
- After the allotted presentation time has expired, competitors will have a fifteen (15) second grace period to conclude their presentation before they are cut-off to begin the question period.
- Time to both ask and answer questions will be counted during the question period.

1.6.5. Presentation Order

- Presentation order shall be determined randomly
- Presentation order shall be announced thirty (30) minutes before the presentations commence. All teams are required to be present at this announcement
- Teams are not allowed to switch places in the presentation order
- Teams may not be in the audience for presentations of their competition until after they have completed their presentation. Any attempts at entering the presentation room before their allotted time will result in a warning for a first offense, and a ten (10) point penalty for any further offenses.

1.6.6. Deliverables

1.6.6.1. For Competitors

Prior to the competition, information about the rules of the competition will be available on the competition website, metengcomp.com. Six (6) days before the competition, registered teams will receive a broad biological research topic to prepare and submit a pre-competition literature review. The specific Bio-Engineering problem, requirements, and constraints will be provided on Day 1 of the competition.

1.6.6.2. From Competitors

Each team must electronically submit a written technical report before the end of the eight (8) hour design period. The report must not exceed six (6) pages of main content, excluding the title page and references. Any supporting appendices must not be used to bypass the six-page limit. Teams must also develop a non-physical engineering prototype or design representation demonstrating their proposed solution. Physical prototypes are not permitted.

The report and non-physical prototype will be made available to the judges before each team's presentation. Each team is required to prepare a presentation explaining their proposed solution, including the relevant biological principles, engineering design, technical feasibility, and implementation. All deliverables must be completed and submitted before the end of the allotted design period.

1.6.7. Response to Questions

- Only the Competition Lead(s) may answer questions during the problem presentation and design phases. Volunteers and judges may not answer questions regarding rules and procedures at any time.
- During the presentation of the problem, the Competition Leads shall answer orally and write down the answers provided.
- During the design phase, only questions related to deliverable content shall be answered. No answers shall be provided in response to questions about the problem that might lead to the development of a new approach or which might invalidate a solution.
- During the design phase, answers to questions shall be provided in writing to all teams at the same time. Fifteen (15) minutes may be allowed to elapse for a significant number of questions accumulated before responses are provided. A copy of the responses must be provided to the judges prior to the presentations.

1.7. ASSESSMENT AND JUDGING

- All competitions must have a minimum of three (3) judges (and in any excess, an odd number of judges).
- Judges are required to assess the problem-solving abilities, proposed solution, and communication skills of the competitors.
- Judges for Bio-Engineering should come from a variety of backgrounds, including biomedical engineering, biological sciences, technical engineering, and related fields.
- Judges should assess the technical feasibility, biological relevance, innovation, and real-world applicability of the proposed solutions, including the non-physical prototypes.
- Judging feedback shall be provided to each competing team within a month following the conclusion of MEC.

1.7.1. Judging Matrix

Solution	• Clear identification and understanding of the problem • Application of relevant biological concepts and processes • Engineering design and integration of biological components • Smart infrastructure, monitoring, and system control • Technical feasibility, cost, safety, and scalability • Innovation, originality, and practicality of the solution	/60
Report	• Organization, clarity, formatting, and proper citations • Biological research, technical data, calculations, and supporting diagrams • Justification of design decisions and engineering trade-offs	/15
Presentation	• Clear, structured, and professional presentation of the proposed solution • Effective use of non-physical prototypes, engineering representations, and visual aids • Technical understanding and ability to answer judges' questions • Team participation, communication, and time management	/25
Penalties	• Plagiarism • Insufficient Citation • Documents Received After Deadline • Absent Team Member	-50 -50 -25 -25
Total		/100